

Projected (w/o VC)	refinery overhead	\$	4,500.00			Direct Labor	\$	600.00			Direct Labor	\$	300.00			Total	\$	375.00	TVC/lb	\$	0.25							
	Variable Costs					Total	\$	600.00	TVC/lb	\$	0.20	Total	\$	1,393.75	TVC/lb	\$	0.46	Gross Profit	\$	(375.00)	GP/lb	\$	(0.25)					
	X	\$	-	X/lb	1.333333	Gross Profit	\$	1,800.00	GP/lb	\$	0.60	Gross Profit	\$	556.25	GP/lb	\$	0.19											
	Y	\$	-	Y/lb	0.729167	The only meaningful discussion here is if we proceed under the preposition that the costs for 7500 units of X and 9000 units of Y has been imbedded in some other account as if it hasn't and is being accounted for as a cost incurred within the upcoming month than we are screwed as that is not enough materials to even produce the primary product at the breakeven point. Using the previous month for its baseline unit costs and profits, we can see that past the breakeven point, the PP still has the highest profit contribution or contribution margin. Hence, if we proceed with that in mind and prioritize PP's production, we will be left with 3000 units of X (as 1 unit of X and 2 units of Y is needed for every lb of PP). With the costs already paid for, it is more profitable to use 1500 units of X to produce 3000 units of K (with a contribution margin of \$0.60) than to buy 1500 units of Y to produce 3000 units of M (with a contribution margin of \$0.19). Hence, there would also be no need to consider treatment and we would still have 1500 units of K to be used the month after.																						
	Direct Labor	\$	1,875.00	DL/lb	\$	0.42																						
	Total	\$	19,375.00	TVC/lb	\$	4.31																						
	Gross Profit	\$	6,275.00	GP/lb	\$	1.39																						
	Total Revenue	\$	28,050.00																									
	Total GP	\$	8,075.00																									
	Cost Benefit Analysis of Primary Product					Cost Benefit Analysis of M																	Cost Benefit Analysis of Treatment					Given
Revenue	\$	42,750.00	Amt of PP	7500	Revenue	\$	3,250.00	Waste	2500	Revenue	\$	-	Waste	2500	X	7500												
Fixed Costs					Variable Costs	\$		M / Waste	1	Variable Costs	\$		T / Waste	1	Y	9000												
Administrative expenses	\$	13,000.00			Y	\$	1,822.92	Amt of M	5000	Direct Labor	\$	625.00	Amt to T	2500														
refinery overhead	\$	4,500.00			Direct Labor	\$	500.00			Total	\$	625.00	TVC/lb	\$	0.25													
Variable Costs					Total	\$	2,322.92	TVC/lb	\$	0.46	Gross Profit	\$	(625.00)	GP/lb	\$	(0.25)												
X	\$	-	X/lb	1.333333	Gross Profit	\$	927.08	GP/lb	\$	0.19																		
Y	\$	4,375.00	Y/lb	0.729167	As we can see here, buying additional Y (a total of 6000 *Y for P and 2500 *Y for M) is the optimal move here to maximize Gross Profit as 46000 is greater than 27600																							
Direct Labor	\$	3,125.00	DL/lb	\$	0.42																							
Total	\$	25,000.00	TVC/lb	\$	3.33																							
Gross Profit	\$	17,750.00	GP/lb	\$	2.37																							
Total Revenue	\$	46,000.00																										
Total GP	\$	18,677.08																										
w/o K but buy Y																												
w/o K, not buy Y																												
Total Revenue		27600																										
Total GP		6831.25																										

Question 1.

Using Python

```
!pip install pulp
from pulp import LpMaximize, LpProblem, LpVariable

# Define the LP problem
model = LpProblem("Chemical_Manufacturing_Optimization", LpMaximize)

# Decision variables
P = LpVariable("Primary_Product", lowBound=0, cat="Continuous")
K = LpVariable("Product_K", lowBound=0, cat="Continuous")
M = LpVariable("Product_M", lowBound=0, cat="Continuous")
T = LpVariable("Waste_Treated", lowBound=0, cat="Continuous")

# Objective function: Maximize profit
model += (
    5.70 * P + 0.80 * K + 0.65 * M - 0.25 * T
    - 0.20 * K - 0.10 * M,
    "Total_Profit"
)

# Constraints
model += P + K <= 7500, "Material_X_Limit"
model += 2 * P + M <= 9000, "Material_Y_Limit"
model += P == K + M + T, "Waste_Balance"

# Solve the model
model.solve()

# Print results
print("Optimal Production Plan:")
print(f"Primary Product (P): {P.varValue} lbs")
print(f"Product K (K): {K.varValue} lbs")
print(f"Product M (M): {M.varValue} lbs")
print(f"Waste Treated (T): {T.varValue} lbs")
print(f"Total Profit: ${model.objective.value():.2f}")
```

```
Collecting pulp
  Downloading PuLP-3.0.2-py3-none-any.whl.metadata (6.7 kB)
Downloaded PuLP-3.0.2-py3-none-any.whl (17.7 MB)
17.7/17.7 MB 84.9 MB/s eta 0:00:00
Installing collected packages: pulp
Successfully installed pulp-3.0.2
Optimal Production Plan:
Primary Product (P): 4500.0 lbs
Product K (K): 3000.0 lbs
Product M (M): 0.0 lbs
Waste Treated (T): 1500.0 lbs
Total Profit: $27,075.00
```

Using Excel. To clarify we used the “solver” tool in the Data tab in excel to find the optimal solution to the problems given.

Microsoft Excel 16.0 Answer Report

Worksheet: [T1_TMP_chem.xlsx]Sheet2

Report Created: 2/23/2025 5:08:25 PM

Result: Solver found a solution. All Constraints and optimality conditions are satisfied.

Solver Engine

Engine: Simplex LP

Solution Time: 0.047 Seconds.

Iterations: 4 Subproblems: 0

Solver Options

Max Time Unlimited, Iterations Unlimited, Precision 0.000001

Max Subproblems Unlimited, Max Integer Sols Unlimited, Integer Tolerance 1%, Assume NonNegative

Objective Cell (Max)

Cell	Name	Original Value	Final Value
\$B\$5	Total Profit	0	27075

Variable Cells

Cell	Name	Original Value	Final Value	Integer
\$B\$1	Primary Product (PP)	0	4500	Contin
\$B\$2	Product K (K)	0	3000	Contin
\$B\$3	Product M (M)	0	0	Contin
\$B\$4	Waste Treated (T)	0	1500	Contin

Constraints

Cell	Name	Cell Value	Formula	Status	Slack
\$D\$1	Primary Product (PP)	7500	\$D\$1<=7500	Binding	0
\$D\$2	Product K (K)	9000	\$D\$2<=9000	Binding	0
\$D\$3	Product M (M)	0	\$D\$3=0	Binding	0
\$B\$1	Primary Product (PP)	4500	\$B\$1>=0	Not Binding	4500
\$B\$2	Product K (K)	3000	\$B\$2>=0	Not Binding	3000
\$B\$3	Product M (M)	0	\$B\$3>=0	Binding	0
\$B\$4	Waste Treated (T)	1500	\$B\$4>=0	Not Binding	1500

First the variables p, k, m, and t are defined as shown. Then using the formula with the given information in the scenario we get the following to answer question 1:

Value of Additional Pounds of Each Raw Material The shadow prices indicate the potential increase in total profit if the company could acquire more raw materials:

1. Material X: The shadow price is \$0.85, meaning each additional pound of Material X would increase total profit by \$0.85, assuming all else remains constant
2. b. Material Y: The shadow price is \$2.40, which is significantly higher than Material X.
3. This suggests that Material Y is the more constrained and valuable resource—securing, additional Material Y would have a greater impact on improving profitability. This makes the primary product still the most profitable and with the 7500 of X and 9000 of Y, we have a surplus of X compared to a deficit of Y.
4. Waste Balance: The shadow price of \$0.25 indicates that each additional unit of waste that can be disposed of or treated effectively contributes \$0.25 to total profit.
Recommendation: If possible, Smith-Brown Industrial Chemicals Inc. should prioritize securing additional Material Y over Material X, as it has a higher profit contribution per unit.

Question 2.

Python

```
# Sensitivity Analysis: Varying Parameters
price_changes = [5.50, 5.70, 5.90] # Test different selling prices of primary product
profit_results = {}

for price in price_changes:
    # Rebuild the objective function with the new price
    model.setObjective(
        price * P + 0.80 * K + 0.65 * M - 0.25 * T - 0.20 * K - 0.10 * M
    )
    model.solve() # Re-solve the model
    profit_results[price] = model.objective.value()

# Print Sensitivity Analysis Results
print("\nSensitivity Analysis on Primary Product Price:")
for price, profit in profit_results.items():
    print(f"Price: ${price:.2f} -> Total Profit: ${profit:,.2f}")

# Shadow Prices (Constraint Sensitivity)
print("\nShadow Prices (Dual Values for Constraints):")
for name, constraint in model.constraints.items():
    print(f"{name}: {constraint.pi:.2f} (Shadow Price)")

# Reduced Costs (Cost Sensitivity)
print("\nReduced Costs (Impact of Non-Binding Variables):")
for var in [P, K, M, T]:
    print(f"{var.name}: {var.dj:.2f} (Reduced Cost)")
```

Sensitivity Analysis on Primary Product Price:

Price: \$5.50 -> Total Profit: \$26,175.00

Price: \$5.70 -> Total Profit: \$27,075.00

Price: \$5.90 -> Total Profit: \$27,975.00

Shadow Prices (Dual Values for Constraints):

Material_X_Limit: 0.85 (Shadow Price)

Material_Y_Limit: 2.40 (Shadow Price)

Waste_Balance: 0.25 (Shadow Price)

Reduced Costs (Impact of Non-Binding Variables):

Primary_Product: -0.00 (Reduced Cost)

Product_K: -0.00 (Reduced Cost)

Product_M: -1.60 (Reduced Cost)

Waste_Treated: -0.00 (Reduced Cost)

From Excel

	X	\$	-	X/lb	1.333333
	Y	\$	4,375.00	Y/lb	0.729167
	Direct Labor	\$	3,125.00	DL/lb	\$ 0.42
	Total	\$	25,000.00	TVC/lb	\$ 3.33
	Gross Profit	\$	17,750.00	GP/lb	\$ 2.37
w/oK but buy Y	Total Revenue	\$	46,000.00		
	Total GP	\$	18,677.08		
w/o K, not buy Y	Total Revenue		27600		
	Total GP		6831.25		

As we can see here, buying additional Y (a total of $6000 \cdot Y$ for P and $2500 \cdot Y$ for M) is the optimal move here to maximize Gross Profit as 46000 is greater than 27600

Question 3.

The value of additional pounds of each raw material, ergo, X and Y is very different because of the ratio of what is given and the ratio of what is needed/ used for the creation of primary product and the other subsequent materials.

The only meaningful discussion here is if we proceed under the preposition that the costs for 7500 units of X and 9000 units of Y has been imbibed in some other account as if it hasn't and is being accounted for as a cost incurred within the upcoming month than we are screwed as that is not enough materials to even produce the primary product at the breakeven point. Using the previous month for its baseline unit costs and profits, we can see that past the breakeven point, the PP still had the highest profit contribution or contribution margin. Hence, if we proceed with that in mind and prioritize PP's production, we will be left with 3000 units of X (as 1 unit of X and 2 units of Y is needed for every lb of PP). With the costs already paid for, it is more profitable to use 1500 units of X to produce 3000 units of K (with a contribution margin of \$0.60) than to buy 1500 units of Y to produce 3000 units of M (with a contribution margin of \$0.19). Hence, there would also be no need to consider treatment, and we would still have 1500 units of K to be used the month after.

The only meaningful discussion here is if we proceed under the preposition that the costs for 7500 units of X and 9000 units of Y has been imbibed in some other account as if it hasn't and is being accounted for as a cost incurred within the upcoming month than we are screwed as that is not enough materials to even produce the primary product at the breakeven point. Using the previous month for its baseline unit costs and profits, we can see that past the breakeven point, the PP still has the highest profit contribution or contribution margin. Hence, if we proceed with that in mind and prioritize PP's production, we will be left with 3000 units of X (as 1 unit of X and 2 units of Y is needed for every lb of PP). With the costs already paid for, it is more profitable to use 1500 units of X to produce 3000 units of K (with a contribution margin of \$0.60) than to buy 1500 units of Y to produce 3000 units of M (with a contribution margin of \$0.19). Hence, there would also be no need to consider treatment and we would still have 1500 units of K to be used the month after.

This answer came from the first table which is provided in excel.

Projected (w/o VC)	refinery overhead	\$	4,500.00			Direct Labor	\$	600.00			Direct Labor	\$	300.00			Total	\$	375.00	TVC/lb	\$	0.25		
	Variable Costs					Total	\$	600.00	TVC/lb	\$	0.20	Total	\$	1,393.75	TVC/lb	\$	0.46	Gross Profit	\$	(375.00)	GP/lb	\$	(0.25)
	X	\$	-	X/lb	1.333333	Gross Profit	\$	1,800.00	GP/lb	\$	0.60	Gross Profit	\$	556.25	GP/lb	\$	0.19						
	Y	\$	-	Y/lb	0.729167	The only meaningful discussion here is if we proceed under the preposition that the costs for 7500 units of X and 9000 units of Y has been imbibed in some other account as if it hasn't and is being accounted for as a cost incurred within the upcoming month than we are screwed as that is not enough materials to even produce the primary product at the breakeven point. Using the previous month for its baseline unit costs and profits, we can see that past the breakeven point, the PP still has the highest profit contribution or contribution margin. Hence, if we proceed with that in mind and prioritize PP's production, we will be left with 3000 units of X (as 1 unit of X and 2 units of Y is needed for every lb of PP). With the costs already paid for, it is more profitable to use 1500 units of X to produce 3000 units of K (with a contribution margin of \$0.60) than to buy 1500 units of Y to produce 3000 units of M (with a contribution margin of \$0.19). Hence, there would also be no need to consider treatment and we would still have 1500 units of K to be used the month after.																	
	Direct Labor	\$	1,875.00	DL/lb	\$	0.42																	
	Total	\$	19,375.00	TVC/lb	\$	4.31																	
	Gross Profit	\$	6,275.00	GP/lb	\$	1.39																	
	Total Revenue	\$	28,050.00																				
	Total GP	\$	8,075.00																				
Cost Benefit Analysis of Primary Product						Cost Benefit Analysis of M				Cost Benefit Analysis of Treatment				Given									
Revenue	\$	42,750.00	Amt of PP	7500	Revenue	\$	3,250.00	Waste	2500	Revenue	\$	-	Waste	2500	X	7500							
Fixed Costs					Variable Costs			M / Waste	1	Variable Costs			T / Waste	1	Y	9000							
Administrative expenses	\$	13,000.00			Y	\$	1,822.92	Amt of M	5000	Direct Labor	\$	625.00	Amt to T	2500									
refinery overhead	\$	4,500.00			Direct Labor	\$	500.00			Total	\$	625.00	TVC/lb	\$	0.25								
Variable Costs					Total	\$	2,322.92	TVC/lb	\$	0.46	Gross Profit	\$	(625.00)	GP/lb	\$	(0.25)							
X	\$	-	X/lb	1.333333	Gross Profit	\$	927.08	GP/lb	\$	0.19													
Y	\$	4,375.00	Y/lb	0.729167	As we can see here, buying additional Y (a total of 6000*Y for P and 2500*Y for M) is the optimal move here to maximize Gross Profit as 46000 is greater than 27600																		
Direct Labor	\$	3,125.00	DL/lb	\$	0.42																		
Total	\$	25,000.00	TVC/lb	\$	3.33																		
Gross Profit	\$	17,750.00	GP/lb	\$	2.37																		
Total Revenue	\$	46,000.00																					
Total GP	\$	18,677.08																					
w/oK but buy Y		27600																					
w/o K, not buy Y		6831.25																					
Total GP																							

Question 4.

Excel

Microsoft Excel 16.0 Sensitivity Report
Worksheet: [T1_TMP_chem.xlsx]Sheet2
Report Created: 2/23/2025 5:08:26 PM



Variable Cells

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
\$B\$1	Primary Product (PP)	4500	0	5.7	1E+30	3
\$B\$2	Product K (K)	3000	0	0.6	3	0.85
\$B\$3	Product M (M)	0	-1.5	0.55	1.5	1E+30
\$B\$4	Waste Treated (T)	1500	0	-0.25	0.85	0.75

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
\$D\$1	Primary Product (PP)	7500	0.85	7500	1500	3000
\$D\$2	Product K (K)	9000	2.3	9000	6000	1500
\$D\$3	Product M (M)	0	0.25	0	1500	1E+30

With limit report

Microsoft Excel 16.0 Limits Report
Worksheet: [T1_TMP_chem.xlsx]Sheet2
Report Created: 2/23/2025 5:08:27 PM

Objective		
Cell	Name	Value
\$B\$5	Total Profit	27075

Variable			Lower		Upper	
Cell	Name	Value	Limit	Objective Result	Limit	Objective Result
\$B\$1	Primary Product (PP)	4500	4500	27075	4500	27075
\$B\$2	Product K (K)	3000	3000	27075	3000	27075
\$B\$3	Product M (M)	0	0	27075	0	27075
\$B\$4	Waste Treated (T)	1500	1500	27075	1500	27075

Second Sensitivity Analysis

Microsoft Excel 16.0 Sensitivity Report
Worksheet: [T1_TMP_chem.xlsx]Sheet1
Report Created: 2/25/2025 1:15:42 PM

Variable Cells

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
\$B\$1	PP	4500	0	5.7	1E+30	3.85
\$B\$2	M	0	-1.925	0.55	1.925	1E+30
\$B\$3	T	4500	0	-0.25	1E+30	1.283333333
\$B\$5	X	0	0	0	0	1E+30
\$B\$6	Y	0	0	0	0	1E+30

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
\$C\$1	PP	4500	0	7500	1E+30	3000
\$C\$2	M	9000	2.725	9000	6000	9000
\$C\$3	T	0	0.25	0	4500	1E+30

Sensitivity Report

Microsoft Excel 16.0 Sensitivity Report
Worksheet: [T1_TMP_chem.xlsx]Sheet1
Report Created: 2/25/2025 1:15:42 PM

Variable Cells

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
\$B\$1	PP	4500	0	5.7	1E+30	3.85
\$B\$2	M	0	-1.925	0.55	1.925	1E+30
\$B\$3	T	4500	0	-0.25	1E+30	1.283333333
\$B\$5	X	0	0	0	0	1E+30
\$B\$6	Y	0	0	0	0	1E+30

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
\$C\$1	PP	4500	0	7500	1E+30	3000
\$C\$2	M	9000	2.725	9000	6000	9000
\$C\$3	T	0	0.25	0	4500	1E+30

PP	4500	4500
M	0	9000
T	4500	0
Profit	24525	
X	0	
Y	0	
Waste		

Outcome

Microsoft Excel 16.0 Answer Report

Worksheet: [T1_TMP_chem.xlsx]Sheet1

Report Created: 2/25/2025 1:15:41 PM

Result: Solver found a solution. All Constraints and optimality conditions are satisfied.

Solver Engine

Engine: Simplex LP

Solution Time: 0.031 Seconds.

Iterations: 3 Subproblems: 0

Solver Options

Max Time Unlimited, Iterations Unlimited, Precision 0.000001, Use Automatic Scaling

Max Subproblems Unlimited, Max Integer Sols Unlimited, Integer Tolerance 1%, Assume NonNegative

Objective Cell (Max)

Cell	Name	Original Value	Final Value
\$B\$4	Profit	0	24525

Variable Cells

Cell	Name	Original Value	Final Value	Integer
\$B\$1	PP	0	4500	Contin
\$B\$2	M	0	0	Contin
\$B\$3	T	0	4500	Contin
\$B\$5	X	0	0	Contin
\$B\$6	Y	0	0	Contin

Constraints

Cell	Name	Cell Value	Formula	Status	Slack
\$C\$1	PP	4500	\$C\$1<=7500	Not Binding	3000
\$C\$2	M	9000	\$C\$2<=9000	Binding	0
\$C\$3	T	0	\$C\$3=0	Binding	0
\$B\$1	PP	4500	\$B\$1>=0	Not Binding	4500
\$B\$2	M	0	\$B\$2>=0	Binding	0
\$B\$3	T	4500	\$B\$3>=0	Not Binding	4500

Sensitivity Analysis of Objective Function Coefficients. The sensitivity analysis on the primary product price suggests that profit is highly responsive to changes in the selling price of the primary product:

- If the price drops to \$5.50, profit decreases to \$26,175.
- If the price increases to \$5.90, profit increases to \$27,975.

- c. This highlights that pricing strategy plays a crucial role in overall profitability.
- d. Additionally, the Reduced Costs reveal the following insights:
 - i. Product M has a reduced cost of $-\$1.60$, meaning that for it to become part of the optimal solution, its price must increase by at least $\$1.60$ per pound to break even.
 - ii. Product K has a reduced cost of 0, suggesting that it is part of the current optimal solution (or very close to being profitable).

Question 5.

```
!pip install pulp
from pulp import LpMaximize, LpProblem, LpVariable

# Define the LP problem
model = LpProblem("Chemical_Manufacturing_Optimization", LpMaximize)

# Decision variables
P = LpVariable("Primary_Product", lowBound=0, cat="Continuous")
M = LpVariable("Product_M", lowBound=0, cat="Continuous")
T = LpVariable("Waste_Treated", lowBound=0, cat="Continuous")

# Objective function: Maximize profit
model += (
    5.70 * P + 0.65 * M - 0.25 * T
    - 0.10 * M,
    "Total_Profit"
)

# Constraints
model += P <= 7500, "Material_X_Limit"
model += 2 * P + M <= 9000, "Material_Y_Limit"
model += P == M + T, "Waste_Balance"

# Solve the model
model.solve()

# Print results
print("Optimal Production Plan:")
print(f"Primary Product (P): {P.varValue} lbs")
print(f"Product M (M): {M.varValue} lbs")
print(f"Waste Treated (T): {T.varValue} lbs")
print(f"Total Profit: ${model.objective.value():,.2f}")
```

Collecting pulp

Downloading PuLP-3.0.2-py3-none-any.whl.metadata (6.7 kB)

Downloading PuLP-3.0.2-py3-none-any.whl (17.7 MB)

17.7/17.7 MB 12.7 MB/s eta 0:00:00

Installing collected packages: pulp

Successfully installed pulp-3.0.2

Optimal Production Plan:

Primary Product (P): 4500.0 lbs

Product M (M): 0.0 lbs

Waste Treated (T): 4500.0 lbs

Total Profit: \$24,525.00

5. Accountant's Recommendation on Eliminating Product K The accountant suggested eliminating Product K, believing it to be unprofitable. However, based on the sensitivity analysis:
 - a. Product K has a reduced cost of \$0.00, meaning it is either profitable or very close to the break-even point in the current model.
 - b. Eliminating Product K would force more waste into either special treatment (which incurs a cost of \$0.25 per pound) or production of Product M, which is significantly less profitable due to a -\$1.60 reduced cost. If Product K were eliminated, the optimal solution would likely shift more weight toward:

- i. Producing more Product M, despite its high reduced cost.
- ii. Treating more waste, which would reduce overall profit. Thus, removing Product K may not be the best decision. Instead, a better approach might be to reassess whether Product K's production cost can be reduced, or its selling price increased slightly to improve its profitability. With Solver

The sensitivity analysis indicates that Material Y is the most valuable resource, with a shadow price of \$2.30 per pound, making it the top priority for securing additional supply.

Product K remains viable, as its reduced cost is \$0, meaning it contributes positively to profit, whereas Product M is currently unprofitable with a reduced cost of -\$1.50, requiring a price increase of at least \$2.05 to become viable.

The primary product remains stable, as price fluctuations above \$2.70 do not affect its optimal production. Waste treatment is necessary but should be carefully managed since its cost can rise to \$0.60 before the optimal solution changes.

Eliminating Product K, as suggested by the accountant, would be detrimental, as it would shift more waste into costly treatment or force production of Product M, which is not currently viable. Instead, the company should continue producing Product K, prioritize securing more Material Y, and monitor waste treatment costs carefully to maintain profitability.

Recommendation 1.

Continue Producing Product K – Since Product K has a reduced cost of \$0, it contributes positively to profit. Eliminating it, as suggested by the accountant, would increase waste treatment costs or force the production of Product M, which is currently unprofitable (-\$1.50 reduced cost).

Recommendation 2.

Do Not Produce Product M Unless Its Price Increases – Product M is not part of the optimal solution, and its selling price must increase to at least \$2.05 for it to be viable. 3. Keep Primary Product Pricing Above \$2.70 – The primary product remains stable unless its price drops below \$2.70, at which point profitability will be impacted.

Reflection.

This project required a combination of accounting knowledge, mathematical optimization, and computational tools to determine the most cost-effective method for handling liquid waste in a chemical manufacturing process. I already had experience with accounting principles and equation maximization, including calculus-based derivations to find maxima and minima.

However, to effectively solve this problem, I needed to learn new tools such as PuLP in Python and Excel Solver, both of which are essential for formulating and solving linear optimization problems.

One of the biggest challenges in completing this project was that it did not involve traditional data analysis but instead required applying optimization techniques to an accounting-style problem. Unlike standard data science tasks that focus on large datasets, pattern recognition, or predictive modeling, this project required structuring constraints, defining an objective function, and interpreting sensitivity analysis results. Learning how to properly set up constraints in Excel Solver and formulating an optimization problem in PuLP took time but ultimately helped in achieving a structured and analytical approach to decision-making.

Overall, this experience reinforced my understanding of optimization and decision science while introducing me to new computational tools that will be useful for similar analytical challenges in the future. It also highlighted the importance of adapting data science techniques to real-world business problems, even when they do not involve traditional datasets.